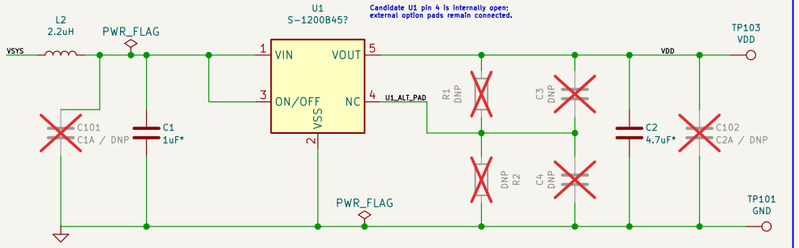
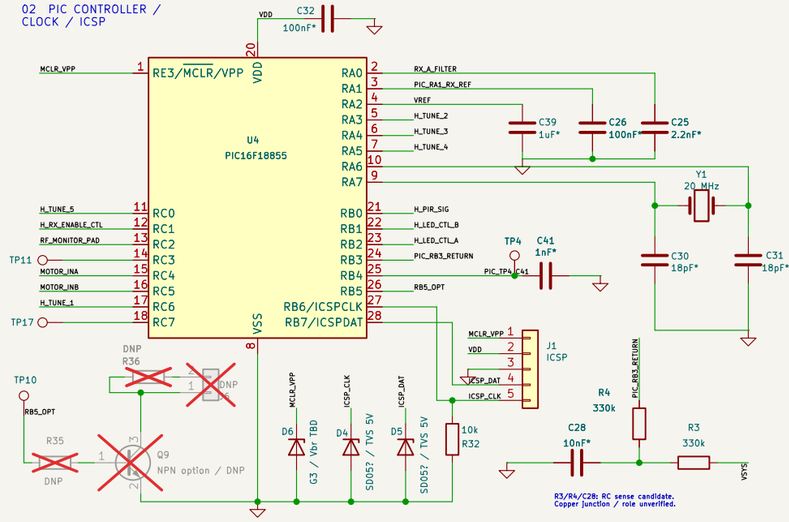
01 POWER / INPUT FILTER

U1: S-1200B45 candidate: VDD measured 4.5 V.



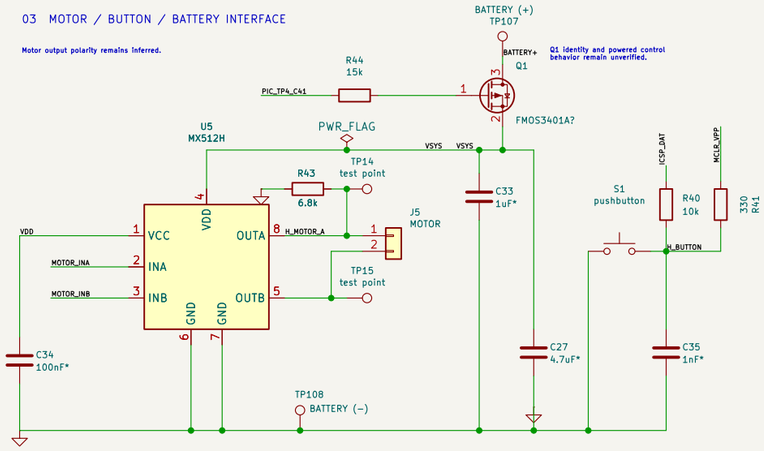
02 PIC CONTROLLER / CLOCK / ICSP



03 MOTOR / BUTTON / BATTERY INTERFACE

Motor output polarity remains inferred.

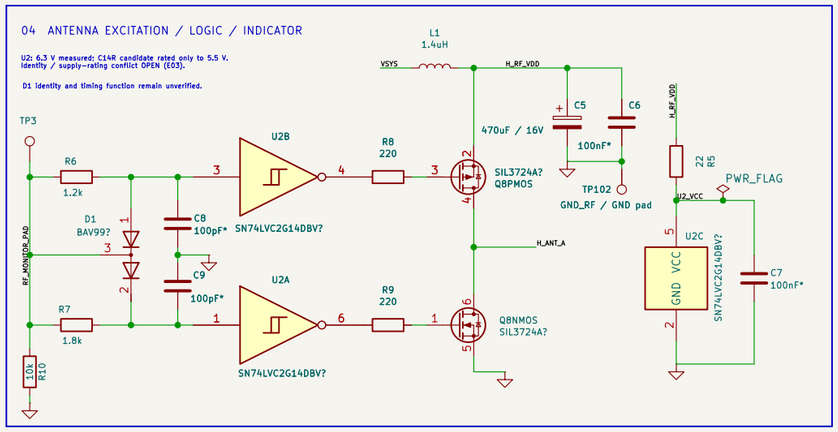
Q1 identity and powered control behavior remain unverified.



04 ANTENNA EXCITATION / LOGIC / INDICATOR

U2: 6.3 V measured; C14R candidate rated only to 5.5 V. Identity / supply-rating conflict OPEN (E03).

D1 identity and timing function remain unverified.



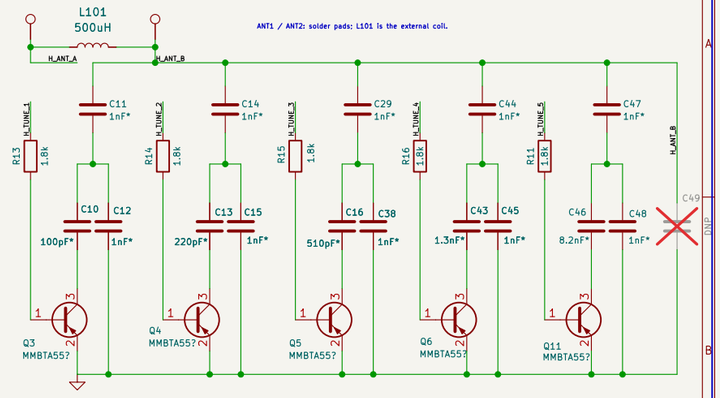
06 SWITCHED ANTENNA CAPACITANCE

Tuning assumes 500 uH and 134.2 kHz; switching weights/order remain unverified.

Q3 diode readings dispute the PNP pinout. Q3/Q4/Q5/Q6/Q11 identities remain unverified.

ANT1 / ANTENNA A

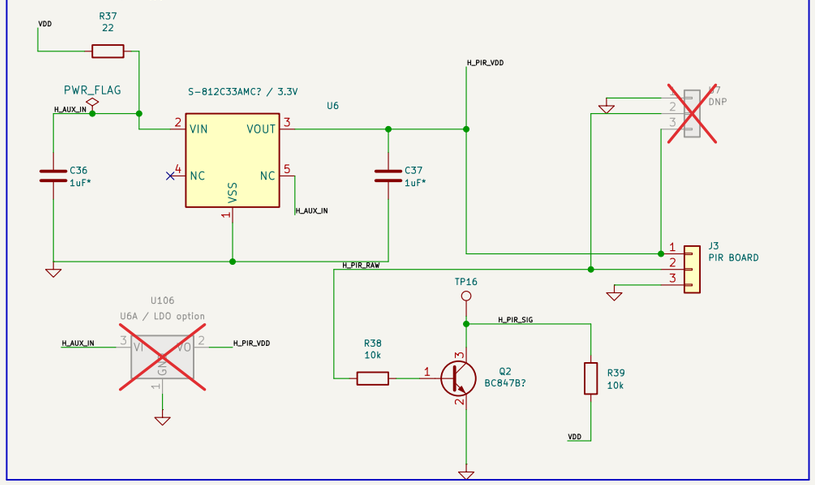
ANT2 / ANTENNA B



07 PIR / AUXILIARY RAIL

U6: S-812C33 candidate; PIR supply measured 3.3 V.

U6A / U106: DNP regulator option. Symbol and ERC conflict retained intentionally.

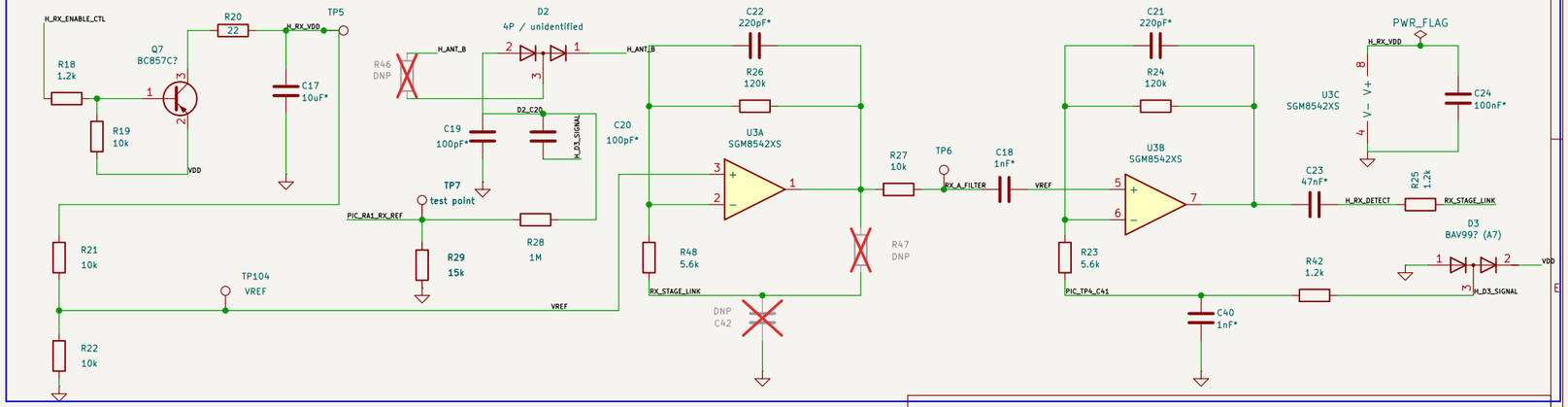


05 RF RECEIVE / GAIN / ENVELOPE

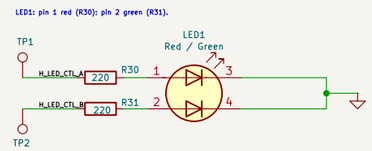
Receiver gain, bias behavior and Q7 switching remain unverified.

C42 / R47 / R46 are unpopulated options.

D2: measured series equivalent; identity unverified.



LED1: pin 1 red (R30); pin 2 green (R31).



RECONSTRUCTION NOTES

Reviewed drawing baseline; not a fabrication-qualified design.

Net labels identify connections; names alone do not establish circuit function.

* = estimated capacitor value; ? = unverified device identity or pin role.

Pad mapping: evidence/pin_crosswalk.json.
Open items: docs/OPEN_ITEMS.md.

TP11 and TP17 are valid endpoints; possible hidden branches remain unproved.

DNP = not fitted. An unresolved route is not an intentional no-connect.

Main board plus external antenna winding, PIR internals, motor and firmware are outside this drawing.

Capacitor estimates and ratings: docs/CAPACITOR_VALUE_REVIEW.html.

v0.9.51 | KiCad 10.0.5 | 2026-10-16

44 starred capacitor estimates; device functions remain qualified.

Owner layout | reviewed reconstruction baseline

Sheet: /
File: PetSafe_1001339.kicad_sch

Title: PetSafe 100-1339 R03 A | single-sheet reconstruction

Size: A3	Date: 2026-10-16	**Rev: 0.9.51**
KiCad E.D.A. 10.0.5		Id: 1/1